IILM UNIVERSITY

INTERNSHIP REPORT

ARTIFICIAL INTELLIGENCE INTERNSHIP

Three Project Portfolio: Rule-Based AI Chatbot, Data Classification Using AI & AI Recommendation Logic

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Program	B.Tech – CSE (AI/ML)
Year	2nd Year
Semester	3rd Semester
Organization	DecodeLabs
Training Batch	2026

Internship Topic / Title: Practical Artificial Intelligence Projects

Academic Year: 2026

CERTIFICATE

This is to certify that **Ameer Hamza**, Enrollment No. **25scs1003004679**, a student of **B.Tech – CSE (AI/ML), 2nd Year, 3rd Semester**, has prepared this internship report based on the Artificial Intelligence industrial training projects assigned by **DecodeLabs** for Batch 2026.

DECLARATION

I, Ameer Hamza, declare that this internship report presents my work on the three assigned Artificial Intelligence projects. The implementations and explanations in this report have been organized for academic and internship submission purposes based on the project briefs supplied by DecodeLabs.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to IILM University and DecodeLabs for providing the opportunity to work on practical Artificial Intelligence projects. The internship tasks helped me understand programming logic, supervised learning, pattern matching, and recommendation concepts through hands-on implementation.

ABSTRACT

This internship report documents three Artificial Intelligence projects completed as part of the DecodeLabs Industrial Training Kit, Batch 2026. Project 1 develops a Rule-Based AI Chatbot using explicit if-else logic and a continuous loop. Project 2 introduces supervised learning through a small classification problem using the Iris dataset and a Decision Tree classifier. Project 3 develops a simple recommendation system that matches user interests with item attributes using similarity logic. Together, the projects demonstrate a progression from programmed decision-making to machine learning and then to personalized recommendation logic.

Keywords: Artificial Intelligence, Python, Rule-Based System, Supervised Learning, Classification, Recommendation System, Similarity, Pattern Matching.

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1. INTRODUCTION

The internship work is organized around three Artificial Intelligence projects supplied in the DecodeLabs Industrial Training Kit. The sequence provides a practical progression: first implementing explicit rules, then training a supervised classification model, and finally building a preference-based recommendation system.

2. INTERNSHIP OBJECTIVES

The main objectives of the internship were to strengthen practical Python programming and develop foundational AI skills through hands-on tasks. Specific objectives were to understand control flow and decision-making, learn the basic supervised-learning pipeline, work with data and model evaluation, and implement preference matching and similarity-based recommendations.

3. PROJECT 1 – RULE-BASED AI CHATBOT

Objective: Create a simple chatbot that responds to predefined user inputs. The supplied project brief requires greetings and exit commands, if-else response logic, and a continuous loop.

Implementation: A Python command-line chatbot was designed using a while loop and if/elif/else conditions. User messages are normalized using strip() and lower(). Predefined rules handle greetings, common questions, help requests, and exit commands. An else branch provides a default response for unsupported input.

Core logic: User input → normalize text → check exit rule → check known rules → return response → repeat.

Example: If the user enters “hello”, the bot returns a greeting. If the user enters “bye”, the bot displays a goodbye message and exits the loop.

Outcome: The project demonstrates control flow, decision-making logic, input handling, and basic rule-based AI concepts.

4. PROJECT 2 – DATA CLASSIFICATION USING AI

Objective: Build a basic classification model using a small dataset. The required workflow is to load and understand a dataset, split it into training and testing sets, and apply a simple classification algorithm.

Dataset: The Iris dataset contains 150 samples, four numerical input features, and three flower classes. It is suitable for demonstrating a compact supervised-learning workflow.

Implementation: The dataset is loaded with scikit-learn, inspected with pandas, and divided into training and test subsets using an 80/20 split. A DecisionTreeClassifier is trained on the training data. Predictions are made on unseen test records and evaluated using accuracy, a classification report, and a confusion matrix.

Result: With a stratified 80/20 split and random_state=42, the completed Decision Tree implementation achieves a high test accuracy on the held-out Iris records. The exact score is reproducible by running the supplied code.

Outcome: The project demonstrates data handling, supervised learning basics, model training, prediction, and evaluation.

5. PROJECT 3 – AI RECOMMENDATION LOGIC

Objective: Create a simple recommendation system based on user preferences. The project requires user input, preference matching through logic or similarity, and display of recommended items.

Implementation: A content-based movie recommendation system was created. Each movie is represented by a set of genres such as action, sci-fi, drama, comedy, or romance. User interests are converted into a set and compared with each movie using Jaccard similarity.

Similarity formula: Jaccard Similarity = |User Preferences ∩ Item Attributes| / |User Preferences ∪ Item Attributes|.

Example: For user preferences {action, sci-fi, thriller}, movies such as The Matrix and Inception receive a perfect 1.00 similarity because all three attributes match.

Outcome: The project demonstrates logic building, pattern matching, personalization, and fundamental recommendation-system concepts.

6. TOOLS AND TECHNOLOGIES

Technology / Tool	Use in Internship
Python	Implementation of all three AI projects
VS Code / Google Colab	Development and execution environment
pandas	Dataset inspection and data handling in Project 2
scikit-learn	Classification model and evaluation in Project 2
Sets / Similarity Logic	Preference matching in Project 3

7. SKILLS AND LEARNING OUTCOMES

The three projects collectively strengthened the following skills:

Project	Primary Skills Developed
Project 1	Control flow, if-else logic, loops, input handling, rule-based decision making
Project 2	Data handling, train/test split, supervised learning, model training, evaluation
Project 3	Logic building, pattern matching, similarity scoring, recommendation concepts

8. PROJECT COMPARISON

Aspect	Project 1	Project 2	Project 3
AI approach	Explicit rules	Supervised learning	Similarity logic
Input	Text message	Feature values	User interests
Main process	If-else decisions	Train/test/model	Preference matching
Output	Chatbot response	Predicted class	Ranked recommendations

9. CHALLENGES AND IMPROVEMENTS

During these projects, the main conceptual challenges are choosing clear rules, structuring data for model training, and selecting meaningful attributes for similarity matching. Future improvements can include expanding the chatbot vocabulary, comparing multiple classification algorithms, testing new data, using weighted recommendation scores, collecting user ratings, and building a graphical or web interface.

10. CONCLUSION

The three-project internship sequence provides a practical foundation in Artificial Intelligence. The first project establishes programmatic decision-making, the second introduces supervised machine learning and model evaluation, and the third demonstrates personalized recommendations through pattern matching. Completing these tasks provides a compact portfolio of foundational AI implementations and creates a base for more advanced projects.

11. REFERENCES / SOURCE MATERIAL

1. DecodeLabs Industrial Training Kit, Artificial Intelligence, Batch 2026 — Project 1: Rule-Based AIChatbot.
2. DecodeLabs Industrial Training Kit, Artificial Intelligence, Batch 2026 — Project 2: Data ClassificationUsing AI.
3. DecodeLabs Industrial Training Kit, Artificial Intelligence, Batch 2026 — Project 3: AI RecommendationLogic.
4. Internship presentation/report-format instructions supplied by the student.

SUBMISSION DETAILS

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